Valid	Invalid
012	05,600(Commas is not allowed)
-027	04.56 (Decimal point is not allowed)**
+0231	0158 (8 is not a permissible digit in octal system)

Notes

** When you use a fractional number that begins with 0, C++ considers the number as an integer not an Octal.

(iii) Hexadecimal

Any sequence of one or more Hexadecimal values (0 …. 9, A …. F) that starts with 0x or 0Xis considered as an Hexadecimal constant.

Valid	Invalid
0x123	0x1,A5 (Commas is not allowed)
0X568	0x.14E (Decimal point is not allowed like this)

The suffix L or l and U or u added with any constant forces it to be represented as a long or unsigned constant respectively.

(2) Real Constants (or) Floating point constants

A real or floating point constant is a numeric constant having a fractional component. These constants may be written in fractional form or in exponent form.

Fractional form of a real constant is a signed or unsigned sequence of digits including a decimal point between the digits. It must have at least one digit before and after a decimal point. It may be prefixed with + or - sign. A real constant without any sign will be considered as positive.

Exponent form of real constants consists of two parts: **(1) Mantissa** and **(2) Exponent.** The mantissa must be either an integer or a real constant. The mantissa followed by a letter E or e and the exponent, should also be an integer.

For example, 58000000.00 may be written as 0.58×10^8 or 0.58E8.

Mantissa (Before E)	Exponent (After E)
0.58	8

Example:

5.864 E1 $10^1 \times 5.864$ ➔ 58.64 ➔

5864 E-2 $10^{-2} \times 5864$ ➔ 58.64 ➔

0.5864 E2 $10^2 \times 0.5864$ ➔ 58.64 ➔

Boolean Literals

Boolean literals are used to represent one of the Boolean values (True or false). Internally true has value 1 and false has value 0.

Character constant

A character constant in C++ is any valid single character enclosed within single quotes.

Valid character constants : 'A', '2', '$'

Invalid character constants : "A"

The value of a single character constant has an equivalent ASCII value. For example, the value of 'A' is 65.

Escape sequences (or) Non-graphic characters

C++ allows certain non-printable characters represented as character constants. Non-printable characters are also called as non-graphic characters. Non-printable characters are those characters that cannot be typed directly from a keyboard during the execution of a program in C++, for example: backspace, tabs etc. These non-printable characters can be represented by using escape sequences. An escape sequence is represented by a backslash followed by one or two characters.

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